Conditional generation of cathode vacancy patterns and improved starting geometries: midterm working draft

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Abstract

We study whether conditional generative models can propose novel, low-energy vacancy configurations in a layered cathode more effectively than simple sampling methods. Our current benchmark uses pristine-derived O3 $Li_nCo_{48}O_{96}$ geometries and machine-learned interatomic potential (MLIP) labels. An energy-conditioned autoregressive conditional variational autoencoder (AR CVAE) improves novel low-tail yield over uniform fixed-count sampling, but a low-energy training-mask mutation baseline remains stronger. A separate displacement CVAE improves atomic starting positions for a fixed-cell optimizer; a deterministic regressor gives similar savings. We report the actual scope of these observations and propose controlled validation using the available data. They do not establish DFT ground states, size transfer, or a CVAE-specific relaxation advantage.

1 Revised objective and motivation

Removing Li from a cathode leaves many possible vacancy arrangements. Even at one composition, different arrangements can have different energies and different responses to relaxation. Directly enumerating and evaluating every arrangement becomes impractical as the number of sites grows. A useful generator would concentrate a fixed evaluation budget on new, physically useful structures rather than repeatedly proposing equivalent configurations. The longer-term application is efficient selection of reference structures for potential improvement and simulations relevant to cathode recycling.

The course question is deliberately narrower: *does a conditional latent generator improve the discovery of distinct low-energy configurations over simple controls?* We also test whether a separate conditional displacement model offers useful optimizer starting points. DFT-based potential fine-tuning and experimental thermodynamic validation are later extensions, not claims of the present benchmark.

An MLIP predicts potential energy from atoms and cell and supplies forces and stress for evaluation or atomistic simulation. A single-point call measures these properties at unchanged coordinates. Geometry relaxation instead moves atoms repeatedly using a chosen optimizer. DFT provides an independently computed electronic-structure reference when coordinates and numerical setup are matched. The newly generated large-cell labels in this work are MLIP predictions, not new DFT calculations.

2 Data and independent units

The main pristine parent is $Li_{48}Co_{48}O_{96}$: 48 possible Li sites and 144 other atoms. A deletion mask with n retained Li therefore has $144 + n$ atoms. Occupancy $o_i = 1$ means Li is present at site i; vacancy masks use $v_i = 1 - o_i$. The main improved-model comparison uses $n \in \{12, 24, 36\}$, or 25%, 50%, and 75% lithiation, in the same parent. The wider available main-parent corpus covers every integer count 0–48.

Parent translations and point operations induce 192 retained site permutations. Masks related by those permutations form one symmetry orbit. The smallest transformed vacancy bit string is the canonical key. Splits and novelty checks group by this key; all frames and teacher labels of a mask stay together. This removes within-parent aliases, not automatically aliases between different supercells. The current networks use a fixed parent site list.

Ricardo to complete: Check the parent description, the short recycling motivation, and the distinction between this MLIP benchmark and the historical/Hu DFT archive. Add the appropriate materials references without implying that their geometries or DFT setups are identical to this deletion benchmark.

3 Conditional occupancy models

At a fixed composition, let E be the total single-point energy in eV and $e = E/(144 + n)$ the energy per atom. Sort the m training energies at count n. The element at one-based index $\lceil 0.2m \rceil$ defines the training low-tail threshold $q_{20}(n)$. The binary label and energy condition are

$$b = \mathbb{1}[e \le q_{20}(n)], \qquad c = (n/48, b). \tag{1}$$

Both classes train the model; the composition-only control receives $n/48$. All thresholds are training-derived and teacher-specific. Requesting $b = 1$ does not enforce an energy inequality or follow an energy gradient.

The encoder $q_\phi(z \mid o, c)$ predicts a diagonal Gaussian with latent dimension eight. The decoder $p_\theta(o \mid z, c)$ receives z, c. Training uses the regularized variational objective

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}[-\log p_\theta(o \mid z, c)] + \beta D_{\mathrm{KL}}(q_\phi(z \mid o, c) \| \mathcal{N}(0, I)) \,. \tag{2}$$

The reconstruction term measures probability assigned to observed masks; KL compares the encoded Gaussian with the prior used for generation. At $\beta = 1$ this is a negative conditional evidence lower bound for the model; $\beta = 0.1$ gives a different regularized objective [2, 3]. Good posterior reconstruction alone does not prove good generation from $z \sim \mathcal{N}(0, I)$.

3.1 Dense likelihoods and count constraints

The legacy dense decoder outputs logits ℓ_i and uses independent Bernoulli reconstruction. Taking the top n logits at generation gives the requested count, but introduces a training–sampling mismatch. The exact-count variant instead uses

$$p(o \mid z, c, \textstyle\sum_i o_i = n) = \frac{\exp(\sum_i \ell_i o_i)}{Z_n(\ell)}, \qquad Z_n(\ell) = \textstyle\sum_{u:\sum_i u_i = n} \exp(\sum_i \ell_i u_i). \tag{3}$$

A dynamic program computes the normalization and samples from this conditional distribution. Count is then enforced in the likelihood and sampling rule.

3.2 Autoregressive decoder

The AR decoder models the occupancy sequence through the exact chain rule,

$$p(o \mid z, c) = \prod_{i=1}^{48} p(o_i \mid o_{<i}, z, c). \tag{4}$$

A GRU with hidden size 64 carries information about earlier choices. Teacher forcing supplies observed earlier occupancies during training; generation feeds back sampled choices. If the remaining count forces a bit to zero or one, that choice has probability one and contributes zero negative log likelihood. The sequence is site order, not physical time or relaxation. Autoregressive normalized sampling also appears in statistical-mechanics generative models [4], although our training objective differs from theirs.

An AR control removes the latent variable and encoder while retaining the stochastic decoder. A composition-only AR CVAE removes the energy class. These controls distinguish contributions of the latent model and energy input from the expressive sequential decoder. Discrete canonicalization handles the retained parent symmetries; the current GRU is not an equivariant atomic graph network.

Beihao to complete: Verify these equations against the portable implementation and one reproduced fit. State all hyperparameters and the actual checkpoint rule. Include reconstruction/KL curves and distinguish fit seeds from latent draws.

4 Adaptation and occupancy evaluation

The original three-count dataset contains 1,536 masks: 1,227 training, 153 validation, and 156 test. Round 0 adds previously evaluated proposals, giving 50,247 masks (40,157/4,967/5,123). Round 1 adds 25,000 fresh development masks, giving 75,247 (60,152/7,493/7,602). Original splits stay protected; new canonical keys receive deterministic hash-based splits. Promoting old assessment data to training means that it is no longer untouched final evidence.

The extra adaptive round warmed from the round-0 AR CVAE, added data, recomputed training low-class thresholds, and balanced training classes. It compared $\beta = 0.1$ and $\beta = 1$. This coupled update is not a pure sample-size experiment. The original AR family was selected on development results before the final pool was scored.

For each method, fit seed, and composition, the final assessment retains 512 attempts. A success must be a distinct orbit absent from frozen history and below the *original* teacher-specific training threshold. The metric is

$$Y = \frac{\#\{\text{distinct novel orbits with } e \le q_{20}^{\text{original}}(n)\}}{\#\{\text{all generation attempts}\}}. \tag{5}$$

Repeated or known proposals remain in the denominator. Frozen history includes all original and development masks. UMA-S-1.2.1 with task `omat` and MACE-MH-1 with head `omat_pbe` scored the same canonical deletion geometries in FP32 with TF32 disabled.

Method	UMA Y (%)	MACE Y (%)
Uniform fixed-count	18.99	19.75
Legacy dense CVAE, $\beta = 0.1$	20.29	23.57
Expanded dense, exact count	27.86	29.58
AR, no latent	25.17	27.08
AR CVAE, composition only	31.88	35.76
AR CVAE, energy condition	**35.57**	**40.06**
Extra adaptive AR, $\beta = 0.1$	33.81	38.91
Extra adaptive AR, $\beta = 1$	27.84	30.43
Mutation, original seed pool	45.40	46.96
Mutation, newest seed pool	52.65	54.08

These means cover three independently trained seeds and three compositions. The energy-conditioned AR CVAE improves on uniform sampling, while low-energy mask mutation remains better. Broader model/data comparisons are complete method comparisons, not isolated causal architecture tests. Mutation from the newest pool also benefits from more labeled history.

Extra fine-tuning did not improve average yield. Assessment thresholds stayed fixed; the decline is not caused by tightening that gate. Stronger KL weighting was associated with small latent KL and lower yield, but neither collapse nor the cause of the coupled update's decline has been isolated. Reconstruction, latent information, and generation are distinct quantities [1]. The $\beta = 0.1$ update found lower individual extremes at some compositions, so best energy and many-success yield should both be reported.

Varad to complete: Recompute the table from stored attempts/energies, add a seed-resolved figure, and check novelty, threshold, missing-label, and cost accounting. Preserve this final pool as a completed experiment.

5 Separate conditional displacement model

Vacancy occupancy leaves all retained atoms at pristine coordinates. To learn improved starting geometries, a separate model predicts displacement $u_{a\alpha}$ in angstroms for each present atom a and Cartesian component α. The fixed parent atom-index frame retains each input orientation; it is not a whole-geometry symmetry canonicalization. Reference fractional sites are mapped through the supplied frame cell; mass-weighted global translation is removed and periodic differences use the minimum-image convention. Missing Li rows are zero padding excluded by a presence mask. The cell is supplied, not predicted.

The fixed-cell UMA corpus contains 1,000 independent masks and 6,000 frames: deletion, four intermediate optimizer frames, and one endpoint for each mask. The split is 787 training, 107 validation, and 106 test masks. A lineage's weights total one. The decoder context contains occupancy (48), Li fraction (1), cohort flag (3), cell deformation (9), and four properties: energy per atom, log-transformed force-component RMS, configurational pressure, and log-transformed deviatoric stress norm. The resulting context has 65 components. Statistics are fitted only to training frames within composition/cohort groups.

Requested endpoint conditions come from attained training endpoints: choose the lowest-energy one among endpoints below the training median force RMS. We do not supply a held-out structure's oracle properties as its desired condition. The encoder reads 576 normalized displacement coordinates and context and predicts a 16-dimensional Gaussian. The decoder predicts coordinates using a presence-masked Gaussian reconstruction objective plus KL. A deterministic regressor receives the same context without a latent variable. Each has one fitted model; three CVAE draws are not three independent fits.

5.1 Observed geometry performance

On 128 fresh vacancy masks, direct geometry-CVAE predictions lowered UMA energy relative to deletion by approximately 19.1 meV/atom (median) and reduced median force-component RMS to about 24% of the deletion value. The regressor gave similar changes. However, no direct output in these groups met maximum force < 0.05 eV/Å. The model therefore predicts improved coordinates, not guaranteed minimized geometries.

On a matched 32-mask UMA fixed-cell optimizer check, all methods converged under the same operational force tolerance. Median steps were 40 from deletion, 24 from the regressor, and 24 from CVAE draw 0. Median wall times in that run were 3.71, 2.19, and 2.19 seconds, respectively. The paired step comparison is more portable than wall time. The measured benefit is a learned-start benefit; a CVAE-specific advantage has not been established.

Force-component RMS and maximum force answer different questions:

$$F_{\mathrm{RMS}} = \sqrt{\frac{1}{3N}\sum_{a,\alpha} F_{a\alpha}^2}, \qquad F_{\max} = \max_a \|F_a\|_2. \tag{6}$$

Here N is the number of present atoms. A small RMS does not guarantee every atom meets the maximum-force threshold. The stated 0.05 eV/Å tolerance is the pilot's operational setting, not a universal standard.

Beihao and Varad to complete: Reproduce the geometry CVAE/regressor on frozen lineages and inspect held-out coordinate errors by composition/cohort. Recompute the stored direct-force and paired-step summaries. Separate posterior reconstruction, prior-decoded physical quality, and optimizer savings. Any newly trained model not yet scored has no measured new force/step result.

6 Validation plan using the current data

The next useful tests isolate model behavior rather than simply adding more calculated structures. For adaptation, compare continued training on round 0 with round-1 training from the same checkpoint at fixed β, training-derived thresholds, validation set, optimizer, and number of updates. Class balancing can be a separate factor. This distinguishes extra data from changes in weighting or optimization. Select on development evidence and state explicitly which historical assessments have been reused.

The wider original main-parent data contain all integer Li counts. A composition-balanced fit and protected interior-count evaluation can test interpolation; an evaluation outside the trained range tests extrapolation. Only exact cached geometries have measured energies. Missing cache entries must remain unknown rather than being dropped from an apparent complete yield. The exactly enumerated 12-site O3 benchmark provides 87 orbit masks across compositions, including 17 at half lithiation. It supports finite-support coverage checks, but requires a compatible separate model. Applying 48-site weights to a padded/truncated vector does not establish size transfer. These prepared labels are MLIP labels, not 87 new DFT calculations.

For vacancy generation, report valid count, orbit uniqueness, novelty relative to frozen history, teacher-specific energy quality, and their combined yield. Best energy versus attempt budget complements low-tail yield. Symmetry-minimized distance to training distinguishes local exploration from broader novelty. Independent fit seeds, not correlated trajectory frames or repeated latent draws, define the main model-replication unit. Include adaptation labels and training cost before claiming computational savings.

For geometry prediction, compare the CVAE against the deterministic regressor using held-out lineages, the same supplied cells, and the same context policy. Check coordinate error, minimum separation, directly measured forces/energies, and paired optimizer steps where existing labels permit. Removing scalar property conditions or comparing endpoint-only with weighted trajectory training are bounded, interpretable ablations. Coordinate error alone does not demonstrate an energy advantage or convergence.

7 Novelty, limitations, and team contributions

The current contribution is a reproducible, symmetry-aware evaluation of conditional vacancy generation with strong local-search controls, together with a separate learned-geometry-start prototype. The useful negative findings are that a stronger model need not beat mutation and that latent stochasticity has not yet improved geometry starts beyond regression. Both guide the next generative-model design choices.

The present data do not prove DFT accuracy, global minima, phase stability, finite-temperature equilibrium, experimental recycling conditions, or transfer between sizes/materials. Different teachers are independent prediction comparisons, not replacements for matched DFT validation. A later DFT audit can select composition-matched CVAE, mutation, random, and teacher-disagreement structures; it is not required to complete the current midterm comparison.

Beihao owns model reproduction, the controlled adaptation study, and the method equations. Varad owns independent metric recomputation, baselines, uncertainty, and figures. Ricardo owns parent/symmetry/provenance checks, materials motivation, and physical interpretation. On November 7 each provides a checked paragraph and result; on November 8 we integrate and freeze tables; on November 9 we verify the approximately five-page body and references. The presentation dates are November 24 and December 1; the final report is due December 12.

All authors to complete: Replace the owner notes with completed contributions, verify author names and course formatting, and select one integrating editor. This editable document is a draft; its scientific statements must remain tied to the archived configurations, source tables, and validation receipts.

References

[1] Alexander A. Alemi, Ben Poole, Ian Fischer, Joshua V. Dillon, Rif A. Saurous, and Kevin Murphy. Fixing a broken ELBO. In *International Conference on Machine Learning*, 2018.

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[3] Kihyuk Sohn, Honglak Lee, and Xinchen Yan. Learning structured output representation using deep conditional generative models. In *Advances in Neural Information Processing Systems*, volume 28, 2015.

[4] Dian Wu, Lei Wang, and Pan Zhang. Solving statistical mechanics using variational autoregressive networks. *Physical Review Letters*, 122:080602, 2019.